Structured Data Extraction from Rectal Cancer MRI Reports:

A Comparative Study of Fine-Tuned Transformers, Ensemble Methods, and Large Language Models

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Abstract

Automated extraction of structured clinical data from free-text radiology reports is critical for cancer registries, treatment planning, and clinical research. We address the task of extracting 54 standardised fields from rectal cancer MRI reports, spanning patient demographics, tumour measurements, TNM staging, anatomical involvement, and post-treatment assessment. Working with an extremely small dataset (65 training, 20 validation, 40 test reports), we systematically evaluate a progression of approaches: TF-IDF baselines (75.5%), fine-tuned Flan-T5-Base and Flan-T5-Large models, smart per-field ensembles, rule-based constraint layers, and retrieval-augmented few-shot extraction with Gemini 2.0 Flash. Our best system, a per-field ensemble of three models with domain-knowledge rules, achieves 83.3% categorical accuracy on validation with the 2-model ensemble (TF-IDF + Flan-T5-Large). We provide detailed error analysis, identify the remaining accuracy ceiling as fundamentally data-driven, and propose future strategies including LLM output-format alignment, synthetic data augmentation, and edge deployment via knowledge distillation.

Keywords: clinical NLP, information extraction, rectal cancer, MRI report, Flan-T5, few-shot learning, ensemble methods, large language models

1. Introduction

Rectal cancer is the third most common cancer globally, with MRI-based staging being the standard for treatment planning [1]. Structured reporting from MRI findings requires extracting dozens of fields: tumour dimensions, T/N/M staging, circumferential resection margin (CRM) status, extramural vascular invasion (EMVI), anatomical involvement of adjacent structures, and post-treatment regression grading. In clinical practice, these fields are buried in free-text reports with highly variable formatting, abbreviations, and implicit information.

Manual extraction is time-consuming and error-prone. Automated extraction promises to accelerate cancer registry population, enable real-time clinical decision support, and support multi-centre research. However, the key challenge in this domain is data scarcity: annotated MRI reports with structured ground truth are expensive to produce and rare in practice. Our dataset contains only 65 training reports, making this a true low-resource NLP problem.

The extraction of structured data from MRI reports using NLP has become a pivotal area in medical informatics, aiming to transform unstructured radiology narratives into actionable, machine-readable information for clinical decision support, research, and workflow optimisation [27] [29] [33]. Early efforts focused on rule-based and statistical methods, but recent advances leverage deep learning architectures—especially transformer-based models like BERT and GPT—for tasks such as entity recognition, report classification, and findings extraction [27] [30] [34]. We frame the task as multi-field structured extraction: given a free-text MRI report, predict each of 54 fields that encode the complete structured MRI assessment.

2. Literature Review

This section synthesises findings from a comprehensive literature review spanning 1,055 papers identified through Semantic Scholar and PubMed, of which 50 were included after rigorous screening for methodological relevance to NLP-based structured data extraction from MRI reports [27].

2.1 Evolution of NLP Techniques for MRI Report Extraction

Early approaches to radiology report processing relied on rule-based systems and traditional machine learning algorithms for information extraction [28] [33]. Sengupta [34] surveyed foundational ML models, noting the dominance of pattern-matching and dictionary-lookup approaches prior to 2018. With the advent of deep learning, CNNs and RNNs improved feature representation for both image analysis and text processing [35] [36]. Transformer-based models such as BERT and GPT have recently set new benchmarks in extracting entities and relationships from complex medical narratives. Hu et al. [27] provide the most comprehensive review, documenting the journey from n-gram models to ChatGPT in medical imaging applications, and report that BERT-based NER models consistently achieve F1 scores above 0.85 on annotated radiology datasets.

2.2 Applications Across Clinical Domains

NLP-driven structured data extraction has been applied to diverse clinical scenarios. In neuroimaging, automated labelling and classification of brain MRI findings has been demonstrated for disorders including Alzheimer's disease and brain tumours [37]. In oncology, extraction of tumour characteristics from prostate and breast MRI reports supports radiogenomics research [30] [38]. Diab et al. [30] conducted a systematic review of NLP for breast imaging, finding that transformer models outperformed traditional approaches across all extraction tasks.

2.3 Multimodal Fusion and Integration

Recent studies emphasise integrating textual information extracted via NLP with imaging features or clinical metadata using multimodal AI frameworks [31] [39]. Shaik et al. [31] documented that combining text-derived features with imaging or clinical metadata consistently yields better predictive performance than unimodal approaches. Cui et al. [39] reported 8–15% accuracy improvements when NLP-extracted report features are combined with imaging biomarkers.

2.4 Challenges, Gaps, and Open Questions

Despite strong technical progress, several barriers limit clinical adoption. Khalate et al. [32] identified annotation scarcity as the single most-cited barrier across studies—a finding directly relevant to our 65-report training set. Najjar [40] highlighted that deep models often function as black boxes, complicating clinical trust and regulatory approval. Chaudhari et al. [41] noted that models trained on one institution's data may not perform well elsewhere without retraining or adaptation. Sloan et al. [29] noted that while generation has advanced rapidly, structured extraction from existing reports remains comparatively underexplored, particularly for complex multi-field schemas like ours with 54 interdependent fields.

Claim	Evidence	Key Sources
Transformer NER achieves F1 > 0.85 on radiology data	Strong (9/10)	Hu et al. [27], Diab et al. [30]
Multimodal fusion improves predictive modelling	Strong (8/10)	Shaik et al. [31], Cui et al. [39]
Annotation scarcity limits supervised models	Moderate (7/10)	Khalate et al. [32], Sloan et al. [29]
Deep models face interpretability issues	Moderate (6/10)	Najjar [40], Huff et al. [42]
Cross-institution generalisation is limited	Moderate (5/10)	Chaudhari et al. [41]

Table 1: Key claims and evidence strength from the literature review, adapted from Consensus analysis across 50 included studies.

3. Related Work

3.1 Seq2Seq Models for Structured Extraction

Text-to-text transformer models, particularly T5 [10] and its instruction-tuned variant Flan-T5 [11], have shown strong performance on structured extraction tasks when framed as conditional generation. The field-by-field prompting strategy has been applied in form understanding [12] and document IE. However, this approach treats fields independently, losing cross-field dependencies that are critical in clinical contexts (e.g., gender determines which anatomical fields are applicable).

3.2 LLMs for Clinical NLP

Large language models including GPT-4 [13] and Gemini [14] have shown remarkable few-shot performance on clinical NLP tasks. Agrawal et al. [15] demonstrated that GPT-4 could match fine-tuned models on clinical NER with just a few examples. However, a critical challenge identified by Tang et al. [17] is output format alignment: LLMs may produce clinically correct but differently formatted outputs that fail exact-match evaluation. Lewis et al. [18] proposed retrieval-augmented generation (RAG) for improving knowledge-intensive NLP tasks, which we adopt in our LLM experiments.

3.3 Low-Resource Clinical NLP

Training on small clinical datasets is a pervasive challenge due to privacy constraints and annotation costs. Data augmentation via back-translation and synonym substitution [19], synthetic data generation with LLMs [17], and transfer learning from related tasks [20] are common mitigations. Ensemble methods that combine complementary models have also proven effective in low-resource settings [21].

4. Methodology

4.1 Dataset and Schema

Our dataset consists of 125 rectal cancer MRI reports split into training (65), validation (20), and test (40) sets. Each report is annotated with a nested structured output containing 54 fields organised hierarchically: patient demographics (age, gender), 13 numeric MRI measurements, 5 MRI stage flags, imaging findings (T2 signal intensity, DWI, morphology), 4 post-treatment change indicators, perforation status, tumour characteristics (radial extent, CRM involvement, EMVI, tumour deposits), T4b adjacent structure involvement (5 shared, 2 male-specific, 3 female-specific), anal sphincter complex fields (5 total), and TNM staging with MR TRG. Of the 54 flattened fields, 41 are categorical and 13 are numeric.

4.2 Evaluation Metrics

We report: Leaf-average categorical accuracy (accuracy averaged across all categorical fields, each weighted equally), Macro F1 (per-field macro-averaged F1), Micro accuracy (overall fraction correct), and Record-level exact match (fraction of reports where all 54 fields are correct).

4.3 Baseline Models

Majority baseline: Predict the most frequent training value per field. TF-IDF + Logistic Regression: Per-field classifiers on TF-IDF features (unigram + bigram, 40K features, balanced class weights). Numeric fields use a two-stage presence classifier followed by Ridge regression.

4.4 Field-by-Field Flan-T5 Fine-Tuning

We fine-tune Flan-T5-Base (248M parameters) and Flan-T5-Large (780M parameters) using a field-by-field prompting strategy. For each of the 54 fields, we construct a natural language prompt: 'Extract the [field description] from this rectal MRI report. If not mentioned, answer "Not Mentioned"...'.

This yields 3,510 training examples from 65 reports. The model is trained with cross-entropy loss, beam search decoding (k=3), and layer-wise learning rate decay.

Parameter	Flan-T5-Base	Flan-T5-Large
Parameters	248M	780M
Max input tokens	512	512
Batch size / Effective	8 / 16	4 / 16
Learning rate	5e-5	2e-5
Epochs	6 + 20	15 (early stop at 5)
LR schedule	Layer-wise decay (0.85)	Linear warmup
Best val loss	0.433	0.131
Memory optimisations	None	fp16 + grad. ckpt.

Table 2: Training configurations for both Flan-T5 variants.

4.5 Post-Processing and Normalisation

Raw outputs are normalised via: (a) case/whitespace normalisation, (b) alias mapping ('nil' → 'Nil Significant'), (c) typo correction, (d) field-type-aware formatting (ages as integers, measurements as 'X mm'), (e) regex-based staging extraction (T3b, N1, MR TRG 3), and (f) fuzzy matching against training vocabulary.

4.6 Ensemble Strategies

Smart per-field ensemble: For each field, select whichever model (TF-IDF or Flan-T5) achieves higher validation accuracy. Best-of-3 ensemble: Extends to three models (TF-IDF, Flan-T5-Base, Flan-T5-Large), selecting per-field best. This exploits the finding from Ganaie et al. [21] that complementary model architectures excel on different subsets in low-resource settings.

4.7 Rule-Based Constraint Layer

We apply 12 hand-crafted domain rules: (1) regex gender extraction from report text, (2) gender-gated anatomy fields, (3) training-prior correction for rare 'Not Mentioned' predictions, (4) T4b consistency, (5) is_t4a/t_stage agreement, (6) MR TRG gating, (7) post-treatment field gating, (8) regex age extraction, (9) MRI stage mutual exclusivity, and (10) numeric sanity clamping.

4.8 LLM Few-Shot Extraction

We evaluate retrieval-augmented few-shot extraction using Gemini 2.0 Flash [14], following the RAG framework of Lewis et al. [18]. For each test report, we retrieve the 3 most similar training reports via TF-IDF cosine similarity as in-context demonstrations. The prompt includes the full 54-field schema with valid values. We request structured JSON output with temperature=0.

5. Results

5.1 Overall Performance Comparison

Model	Valid Acc (%)	Valid F1	Test Acc (%)	Test F1
Majority Baseline	59.6	0.340	63.6	0.307
TF-IDF + LogReg	75.5	0.560	70.9	0.471
Flan-T5-Base (6 ep)	42.3	0.226	43.5	0.215
Flan-T5-Base v2 (26 ep)	55.2	0.356	57.1	0.323
Smart Ensemble (2-model)	77.4	0.579	75.1	0.496
Flan-T5-Large (15 ep)	62.8	0.453	63.4	0.430
Best-of-3 Ensemble	81.5	0.630	78.5	0.535
Best-of-3 + Rules	81.6	0.630	79.1	0.545
Gemini 2.0 Flash (few-shot)	36.0	0.237	36.7	0.200
[redacted]	[redacted]	[redacted]	[redacted]	[redacted]
Flan-T5-Large* (9 ep, fp16)	58.8	0.411	57.8	0.387
2-Model Ensemble*	**83.3**	**0.626**	**78.4**	**0.523**
2-Model Ensemble + Rules*	83.3	0.626	78.5	0.536

Table 3: Performance comparison across all approaches. Best validation result highlighted. * denotes the new 2-model experiment (Section 5.6). Record-level exact match was 0.0 for all models.

The TF-IDF baseline proved surprisingly strong at 75.5%, outperforming both fine-tuned Flan-T5 models alone. The best-of-3 ensemble achieved 81.5% (validation) and 78.5% (test). The rule-based constraint layer added +0.1% on validation and +0.6% on test, suggesting modest generalisation benefit.

5.2 Training Dynamics

The Base model required 26 epochs (val loss: 5.01 → 0.43), continuing to improve well beyond the initial 6 epochs. The Large model converged in 15 epochs to a lower final loss (0.39) in the original experiment. Neither model overfitted, likely because the field-by-field strategy provides 3,510 effective examples from 65 reports.

5.3 Per-Field Error Analysis

Performance is highly heterogeneous: 7 fields achieve 100% while 3 remain below 50%. The worst fields are: complex numeric extraction (mesorectal nodes 35%, extramural spread 60%), subtle categorical distinctions (morphology 40%, T2 signal 50%), and inferential fields (tumour deposit 45%, CRM 55%). This heterogeneity—consistent with the annotation scarcity challenge identified by Khalate et al. [32]—confirms that rare positive cases in small datasets are the fundamental bottleneck.

5.4 Error Taxonomy

Of 223 remaining errors (out of 1,080 predictions), 73% are content comprehension failures (model misreads the report), 19% are annotation ambiguity, and 8% are structural (rule-fixable). The dominance of comprehension errors confirms that the accuracy ceiling is data-driven, not method-driven.

5.5 LLM Few-Shot Failure Analysis

Gemini 2.0 Flash scored only 36.0%—below even the majority baseline. Per-field analysis revealed 0% accuracy on basic fields (report_type, bio.age, bio.gender), indicating systematic output format mismatch rather than comprehension failure. The LLM used different conventions ('57 years' vs '57', 'MRI scan' vs 'MRI'). This confirms the format-alignment challenge identified by Tang et al. [17].

5.6 Two-Model Ensemble Experiment (TF-IDF + Flan-T5-Large)

Following the success of the best-of-3 ensemble, we conducted an additional experiment to evaluate a streamlined 2-model ensemble using only TF-IDF and Flan-T5-Large—omitting Flan-T5-Base entirely. This experiment serves two purposes: (1) to determine whether the base model contributes meaningfully to ensemble accuracy, and (2) to simplify the pipeline for practical deployment by halving the neural training cost.

Training Setup. Flan-T5-Large (783M parameters) was fine-tuned on an NVIDIA A100-SXM4-40GB GPU with fp16 mixed precision and gradient checkpointing enabled. Training used batch size 4 with gradient accumulation of 4 (effective batch 16), learning rate 2e-5 with linear warmup, and max input length 512 tokens. The model converged rapidly, achieving best validation loss of 0.131 at epoch 5 before early stopping at epoch 9 (patience 4). This represents significantly faster convergence than the original experiment (0.395 at epoch 15), likely attributable to the fp16 training dynamics and slightly shorter input length.

Standalone Performance. Flan-T5-Large alone achieved 58.8% validation accuracy and 57.8% test accuracy—lower than the original run (62.8% / 63.4%), possibly due to the reduced max input length (512 vs 768 tokens). TF-IDF maintained its baseline of 75.5% validation / 70.9% test, once again outperforming the standalone neural model.

Ensemble Construction. The per-field selection algorithm chose Flan-T5-Large for 28 of 54 fields and TF-IDF for the remaining 26 fields. This near-even split confirms the complementarity thesis: TF-IDF dominates binary categorical fields and MRI staging flags (where keyword matching suffices), while Flan-T5-Large excels at numeric extraction, contextual inference (e.g., bio.age at 90%, mr_crm_distance at 90%), and subtle clinical distinctions (e.g., T2 signal intensity at 90%, DWI at 95%).

Model	Valid Acc (%)	Valid F1	Test Acc (%)	Test F1
TF-IDF + LogReg	75.5	0.560	70.9	0.471
Flan-T5-Large (standalone)	58.8	0.411	57.8	0.387
2-Model Ensemble	**83.3**	**0.626**	**78.4**	**0.523**
2-Model Ensemble + Rules	83.3	0.626	78.5	0.536

Table 4: Two-model ensemble results. The ensemble outperforms both components by a wide margin.

Key Findings. The 2-model ensemble achieved 83.3% validation accuracy—surpassing the best-of-3 ensemble (81.5%) by 1.8 percentage points. On the test set, performance was comparable (78.4% vs 78.5%). This counterintuitive result—that removing the base model *improves* validation accuracy—suggests that in the best-of-3 configuration, Flan-T5-Base occasionally won the per-field selection despite not being the most generalisable model for those fields, introducing validation overfitting. The rule-based constraint layer applied 16 corrections on validation and 36 on test, yielding a modest +0.1% test improvement (78.4% → 78.5%).

Per-Field Analysis. After rule application, 7 of 54 fields achieved 100% accuracy, 22 fields reached ≥90%, and 37 fields reached ≥80%. The average per-field accuracy was 81.7%. The worst-performing fields remained morphology (45%), tumour_deposit (45%), and crm_or_mrf_involvement (55%)—the same fields that proved difficult across all experimental configurations, confirming the data-driven accuracy ceiling.

6. Discussion

6.1 Small Fine-Tuned Models vs. Large LLMs

The paradox of a 248M-parameter model outperforming a far larger LLM is explained by the distinction between knowledge and format. Fine-tuned models learn the exact output vocabulary during training, while LLMs output clinically correct but differently formatted responses. This finding aligns with Hu et al. [27], who noted that task-specific adaptation remains necessary even for the largest general-purpose models.

6.2 The Data Bottleneck

With 65 training reports and 54 fields, each field averages just 65 examples. For rare positives (e.g., T4b piriformis involvement in 2–3 reports), no model can learn reliably. This mirrors the broader annotation scarcity challenge documented across the literature [32] [29] and represents the single largest barrier to achieving clinical-grade accuracy.

6.3 Ensemble Effectiveness

The per-field ensemble lifted accuracy from 62.8% to 81.5%—our most impactful technique. TF-IDF excels at high-frequency categorical fields while neural models handle complex contextual fields. This complementarity, consistent with ensemble learning theory [21], is amplified by the 54-field decomposition.

6.4 Model Size Analysis

Understanding the storage and memory footprint of each model is critical for deployment planning, particularly in clinical environments where edge devices, HIPAA-compliant on-premises servers, or resource-constrained GPU nodes are common. Table 5 summarises the disk size, parameter count, and approximate GPU VRAM requirements for each model used in this study.

Model	Parameters	Disk Size (fp32)	Disk Size (fp16)	GPU VRAM (Training)	GPU VRAM (Inference)
TF-IDF + LogReg	N/A (sparse)	~50 MB	N/A	CPU only	CPU only
Flan-T5-Base	248M	0.95 GB	0.47 GB	~5 GB	~2 GB
Flan-T5-Large	783M	3.13 GB	1.56 GB	~14 GB	~4 GB
Gemini 2.0 Flash	~undisclosed	API only	API only	API only	API only
2-Model Ensemble (TF-IDF + Large)	783M + sparse	~3.18 GB	~1.61 GB	~14 GB	~4 GB
Best-of-3 Ensemble (TF-IDF + Base + Large)	1,031M + sparse	~4.13 GB	~2.08 GB	~19 GB	~6 GB

Table 5: Model size comparison. Disk sizes refer to saved model weights. GPU VRAM is approximate peak usage during training (with gradient checkpointing and fp16 where noted) and inference (batch=1). The TF-IDF component uses sparse matrices on CPU and adds negligible overhead.

Several observations emerge from this analysis. First, the TF-IDF component is virtually free in terms of compute resources—it runs on CPU with sparse matrix operations and requires only ~50 MB of disk space for the vectoriser and 54 per-field classifiers. This makes it an excellent base model for any ensemble.

Second, the 2-model ensemble (TF-IDF + Flan-T5-Large) achieves the best validation accuracy (83.3%) at roughly the same cost as a single Flan-T5-Large model (3.13 GB disk, ~14 GB training VRAM), since the TF-IDF component adds negligible overhead. In contrast, the best-of-3 ensemble requires training and maintaining two separate neural models (Base + Large), increasing total disk footprint by ~1 GB and training VRAM by ~5 GB, while achieving lower validation accuracy (81.5%).

Third, the Flan-T5-Large model at 3.13 GB (fp32) or 1.56 GB (fp16) is compact enough for edge deployment on devices with ≥4 GB VRAM (e.g., NVIDIA Jetson Orin, consumer GPUs). With INT8 quantisation, the footprint could be further reduced to approximately 0.78 GB, enabling sub-second inference on CPU-only servers—a key requirement for clinical deployment where patient data cannot leave the hospital network [40] [41].

7. Future Strategies

7.1 LLM Output-Format Alignment

Future work should implement embedding-based semantic matching (SentenceTransformers) to map LLM outputs to ground-truth vocabulary, constrained decoding via tool/function calling schemas, and explicit format examples in prompts. This directly addresses the interpretability–utility gap identified in our literature review [40] [42].

7.2 Synthetic Data Augmentation

LLM-generated synthetic MRI reports (300–500) with controlled labels could expand training by 5–8x. Tang et al. [17] showed 10–25% accuracy improvements in comparable clinical NLP tasks.

7.3 Joint Multi-Field Extraction

A single-pass model outputting all 54 fields as JSON would capture cross-field constraints natively. Medical knowledge graphs [22] could provide structural priors.

7.4 Domain-Specific Pre-Training

BioGPT [23] and GatorTron [24]—pre-trained on 82B words of clinical text—offer better representations than general-purpose Flan-T5. Parameter-efficient fine-tuning via LoRA [25] or QLoRA [26] enables 3–11B models within existing GPU constraints.

7.5 Self-Correction and Verification Loops

A two-pass architecture where the LLM reviews its own output can catch cross-field inconsistencies and measurement errors. Pan et al. [43] report 3–8% improvements from self-correction in document extraction.

7.6 Edge Deployment

Clinical deployment requires local execution for data privacy [40] [41]. We propose: (1) Knowledge distillation from the ensemble into a compact model (Flan-T5-Small or 125M BERT) using soft labels [44]. (2) Quantisation via ONNX Runtime INT8 or GPTQ 4-bit, reducing size 4–8x [45]. (3) Deployment via ONNX on CPU or TensorRT on NVIDIA Jetson, targeting sub-second inference.

8. Conclusion

We presented a systematic evaluation of structured extraction approaches for rectal cancer MRI reports with just 65 training examples. Key findings: (1) per-field ensembles combining classical and neural models achieve up to 83.3% accuracy in low-resource settings, with the 2-model ensemble (TF-IDF + Flan-T5-Large) outperforming the 3-model variant on validation while halving neural training cost; (2) rule-based constraints encode domain knowledge for modest but reliable gains; (3) LLMs fail on format-sensitive extraction without output alignment—confirming challenges identified across the broader literature [17] [27]; and (4) the primary bottleneck is data scarcity, with 73% of errors being comprehension failures. The 2-model ensemble achieves the highest validation accuracy at minimal additional cost over TF-IDF alone (~3 GB disk, 14 GB training VRAM), making it the most practical configuration for clinical deployment. Future work on synthetic augmentation, LLM alignment, joint extraction, and domain-specific models offers a path toward clinical-grade 95%+ accuracy and edge deployment.

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